DHOLE PATIL EDUCATION SOCIETY'S
DHOLE PATIL COLLEGE OF ENGINEERING

Department of First year Engineering

Course Code: ESC-102-ELE | **Subject: Basic Electrical Engineering** | **Academic Year – 2025-26**

Question Bank | **Semester -I**

Unit –III AC FUNDAMENTAL	
S.NO:	**QUESTIONS**
Que.1	Explain generation of single-phase sinusoidal voltages and currents, also give their mathematical and graphical representation.
Que.2	What is the phase of an electrical quantity? Explain the term in phase and out of phase of an electrical quantity (Lagging and Leading).
Que.3	Define the following terms. (a) Waveform (b) Cycle (c) Time Period (d) Frequency (e) Instantaneous Value (f) Peak Value/Maximum Value/Amplitude:
Que.4	Define and derive expression for average Value of sinusoidal alternating current in terms of its peak value.
Que.5	Define RMS Value of sinusoidal alternating current in terms of its peak value.
Que.6	Define Form factor and Peak Factor.
Que.7	Prove that the voltage and current in a purely resistive circuit are in phase and derive the expression for power consumed in circuit.
Que.8	A sinusoidally varying a.c. voltage is given by v = 141.4 sin (100 πt) volt. Find its (i) RMS value (ii) average value (iii) frequency
Que.9	Covert polar to rectangular or rectangular to polar form (i) 5 +j10 (ii) 6-j8 (iii) 5 <-36.870
Que.10	Derive the expression for average value of a sinusoidal alternating current in terms of its peak value. Also write the formula for i) Form Factor and ii) Amplitude Factor
Que.11	The rms value of 50 Hz sinusoidal alternating current is 20A. At t=0, its value becomes 10A. Write down the equation for current. Also find the magnitude of current at t=6 ms
Que.12	Explain the concept of phase lag & phase lead by using : 1. mathematical equations 2. waveform 3. and phasor diagram
Que.13	An alternating current varying sinusoidally with a frequency of 50 Hz has an rms value of 10 A. Write the expression for instantaneous value of this current quantity and find its value for i) t = 0.0015 sec ii) t = 0.0075 sec after passing through zero and then increasing negatively
Que.14	If 200V, 50Hz supply is applied across the resistance of 10 ohm, find equation for voltage & current.
Que.15	Obtain the expression for instantaneous power in case of purely inductive circuit and by using this expression, prove that this circuit never consumes power.

Que.16	Prove that the current leads the voltage by 90◦ in a purely capacitive circuit, and derive the expression for the power consumed.
Que.17	A sinusoidally varying alternating voltage of 100 V (rms value) with 50 Hz frequency is applied to a circuit find : 1. The mathematical equation of the voltage; 2. Time Period 3. The instantaneous voltage when t = 1.667 ms; 4. The time when instantaneous voltage is 100 V; 5. Average value of the voltage 6. Maximum value of the voltage.
Que.18	A sinusoidal AC voltage given by v = Vm sinωt is applied across a pure inductor. Obtain the following for this circuit: i) Expression for the instantaneous current ii) Phasor diagram. Iii) waveforms of instantaneous
Que.19	Define admittance of a circuit and hence draw the admittance triangle for a capacitive circuit.
Que.20	An alternating current is given by i=14.14 sin (377 t). Find (a) rms value of the current (b) frequency (c) instantaneous value of the current at t=3 ms (d) time taken by the current to reach 10A for first time after passing through zero
Que.21	Convert each of the following polar phasors into their rectangular form. a) V =100 Vrms ∠60 , b) and V =100 Vrms ∠ - 60
Que.22	Define form factor and peak factor. Root means square value of a pure sine wave shaped alternating current is 10A. Waveform completes half cycle in 10ms when starting at t=0 sec. Write the equation of current and estimate instantaneous values at t=l5ms and t=4ms in first complete cycle.
Que.23	The equation for an alternating current is given by i = 77 sin 314t. Find the peak value, frequency, time period and instantaneous value at t = 2 ms.
Que.24	The equation for an alternating current is given by i = 100 sin(396.8t−30◦) A. Find its r.m.s. value, frequency and phase angle of the current.
Que.25	Determine the following parameters of a voltage v = 200 sin314t. (a) Frequency (b) Form factor (c) Crest factor
Que.26	An alternating Voltage has the Voltage v=282.84sin314t. What are values of (a) frequency (b) RMs Voltage (c) the instantaneous Voltage when t=3mseconds.